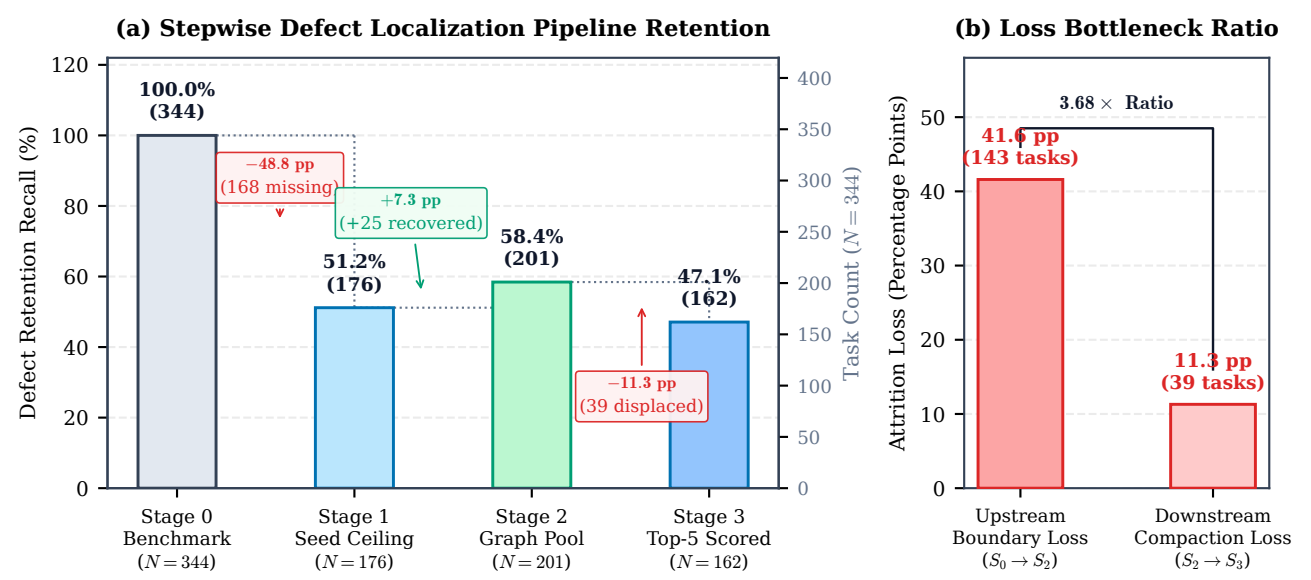
(a) Stepwise Defect Localization Pipeline Retention
Defect Retention Recall (%)
Task Count ($N = 344$)
120
100
80
60
40
20
0
400
350
300
250
200
150
100
50
0
100.0%
(344)
−48.8 pp
(168 missing)
+7.3 pp
(+25 recovered)
51.2%
(176)
58.4%
(201)
47.1%
(162)
−11.3 pp
(39 displaced)
Stage 0
Benchmark
($N = 344$)
Stage 1
Seed Ceiling
($N = 176$)
Stage 2
Graph Pool
($N = 201$)
Stage 3
Top-5 Scored
($N = 162$)
(b) Loss Bottleneck Ratio
Attrition Loss (Percentage Points)
50
40
30
20
10
0
3.68 × Ratio
41.6 pp
(143 tasks)
11.3 pp
(39 tasks)
Upstream
Boundary Loss
($S_0 \to S_2$)
Downstream
Compaction Loss
($S_2 \to S_3$)